



9.5.6 Wrap-Up: Ticket Out the Door

1. Find the solution(s). Show your work.

$$86 = 5x^2 - 34$$

Unit 9, Lesson 6: Solving Quadratic Equations by Completing the Square



Benchmarks

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

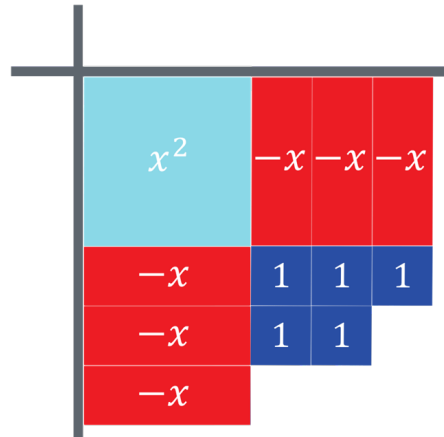
- ☐ I can solve a quadratic by completing the square.
- ☐ I can interpret solutions to quadratics in a real-world context as viable or nonviable.



9.6.1 Warm-Up: Complete the Square

The expression $x^2 - 6x + 5$ is shown.

- How many positive 1-unit tiles should be added to complete the square?



- Adding only the positive unit tiles will completely change the original expression, so how many negative 1-unit tiles should be added?

- Rewrite $x^2 - 6x + 5$ in vertex form by placing a number in each square and a sign in each circle.

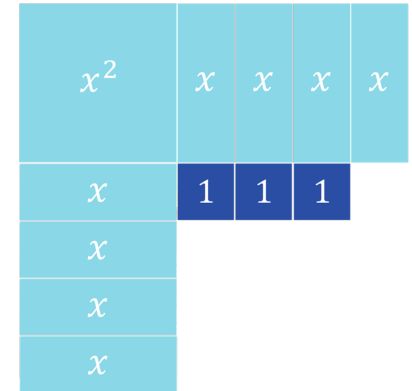
$$(x \bigcirc \square)^2 \bigcirc \square$$



9.6.2 Guided Instruction: Solving Quadratic Equations by Completing the Square

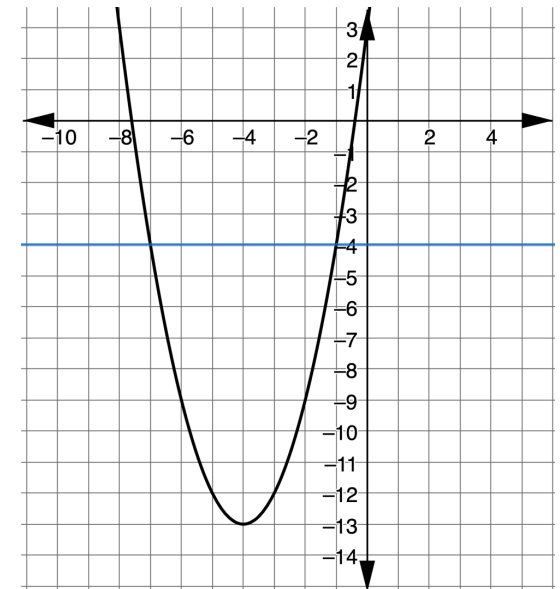
- The expression $x^2 + 8x + 3$ is shown.

Rewrite the expression in vertex form.



- The equations $y = x^2 + 8x + 3$ and $y = -4$ are shown on the coordinate grid.

Where does the line $y = -4$ intersect with the quadratic $y = x^2 + 8x + 3$?



The intersections show the solutions to $x^2 + 8x + 3 = -4$.

3. Daniella tried to solve $x^2 + 8x + 3 = -4$ by using the square root method. Her steps are shown.

$$x^2 + 8x + 3 = -4$$

$$x^2 + 8x = -7$$

$$x^2 = -7 - 8x$$

Daniella got stuck on what to do next. Discuss with your partner why Daniella cannot solve $x^2 = -7 - 8x$ using square roots.

4. One way of solving $x^2 + 8x + 3 = -4$ is by completing the square.

- a. Complete the steps in the table.

Original Equation	$x^2 + 8x + 3 = -4$
Add 4 to both sides to set the equation to 0	$x^2 + 8x + 3 + 4 = -4 + 4$ $x^2 + 8x + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$
Complete the square	$(x^2 + 8x) + 7 = 0$ $(x^2 + 8x + \underline{\hspace{1cm}}) + 7 - \underline{\hspace{1cm}} = 0$ $(x + 4)^2 - 9 = 0$

- b. Solve $(x + 4)^2 - 9 = 0$ by completing the steps.

Add 9 to both sides	$(x + 4)^2 - 9 = 0$ $(x + 4)^2 - 9 + \underline{\hspace{1cm}} = 0 + \underline{\hspace{1cm}}$
Take the square root of both sides	$(x + 4)^2 = 9$ $\sqrt{\underline{\hspace{1cm}}} = \sqrt{\underline{\hspace{1cm}}}$
Because $\sqrt{a^2} = a$	$ x + 4 = \underline{\hspace{1cm}}$
	$x + 4 = 3 \text{ or } x + 4 = -3$
	$x = \underline{\hspace{1cm}} \text{ or } x = \underline{\hspace{1cm}}$

- c. Check your solutions with the intersections found using the graph.

5. Solve $-3(x - 5)^2 + 6 = -21$.

6. Solve by completing the square.

$$-4x^2 - 16x - 11 = -3$$



9.6.3 Guided Practice: Solving Quadratic Equations by Completing the Square

1. Solve by completing the square. Show your work.

$$x^2 - 12x + 7 = -17$$

2. The function $f(x) = -3x^2 + 54x + 83$ models the sales of music cassette tapes for the years 1980 to 2000, in tens of millions of dollars, where x is the number of years since 1980.

- a. A step is shown below in the process of rewriting the function $f(x) = -3x^2 + 54x + 83$. Complete the step.

$$f(x) = -3(x \quad \quad)^2 + 83 \quad \quad$$

- b. The function, f , models the sales in tens of millions of dollars. How many tens of millions of dollars is \$40,000,000?
- c. Solve $-3x^2 + 54x + 83 = 4$ by completing the square.
- d. Determine which solution is not viable.
- e. Justify why that solution is not viable.
- f. In what year were the sales of cassette tapes \$40,000,000?



9.6.4 Your Turn!

Solve by completing the square. Show your work.

1. $-x^2 + 10x + 8 = -6$

2. $6x^2 - 48x - 2 = 4$



9.6.5 Wrap-Up: Growth Mindset

"You may encounter many defeats, but you must not be defeated. In fact, it may be necessary to encounter the defeats, so you can know who you are, what you can rise from, how you can still come out of it." — Maya Angelou

1. What was the toughest part of today's lesson for you?
2. How did you keep going when things got tough?
3. Write a short note to yourself or draw a picture that celebrates your perseverance through the tough part of today's lesson.



Unit 9, Lesson 7: Solving Quadratics by Factoring



Benchmark

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.



Learning Target

- ☐ I can solve a quadratic by factoring.